# 3.1.2 Amount of substance: Sheet 6

*Volumetric techniques · 16 marks · about 25 minutes*

**Key information**

- Standard solution: weigh by difference, dissolve in a beaker, transfer with washings to a volumetric flask, make up so the bottom of the meniscus is on the line, invert to mix.
- Rinse the burette and pipette with the solution they will hold; rinse the conical flask with distilled water only.
- Adding distilled water to the conical flask during a titration does not change the moles of reactant.
- **% uncertainty = uncertainty ÷ measured value × 100**. A burette titre involves two readings; a balance used by difference also involves two readings.
- Bigger titres (eg more dilute solution in the burette) give a smaller percentage uncertainty.

## Section A: Self-quiz

**01** Use this information for Questions 01 and 02.

A student completes a titration to determine the concentration of ethanoic acid in vinegar. 25.0 $cm^3$ of vinegar are transferred to a conical flask using a pipette. A few drops of phenolphthalein are added. Sodium hydroxide solution is added from a burette. The titration is repeated until concordant results are obtained.

Which suggestion improves the accuracy of the titres?

**[1 mark]**

**A** Rinsing the conical flask with vinegar between each titration.
**B** Rinsing the conical flask with sodium hydroxide solution between each titration.
**C** Rinsing the conical flask with water between each titration.
**D** Not rinsing the conical flask between each titration.

**02** Which suggestion decreases the percentage uncertainty in the mean titre?

**[1 mark]**

**A** Use a more dilute solution of sodium hydroxide in the burette.
**B** Use a more dilute solution of vinegar.
**C** Rinse the inside of the conical flask with distilled water during each titration.
**D** Rinse the tip of the burette with distilled water near the end point in each titration.

**03** A student uses the correct method to prepare 250 $cm^3$ of sodium carbonate solution in a volumetric flask.

The uncertainty for the volumetric flask is ±0.20 $cm^3$

Calculate the percentage uncertainty in the volume of this sodium carbonate solution.

**[1 mark]**

**04** A student uses a funnel to fill the burette with sulfuric acid before starting a titration. After filling, the student forgets to remove the funnel from the top of the burette.

Suggest why this might affect the titre volume recorded.

**[1 mark]**

**05** State one advantage of using a conical flask rather than a beaker for a titration.

**[1 mark]**

## Section B: Exam questions

**06** A student uses this method to prepare a standard solution of sodium carbonate.

1. Weigh a clean, dry, empty container on a balance that reads to 2 decimal places.
2. Add about 2.5 g of solid sodium carbonate to the container.
3. Tip the solid into a beaker.
4. Add approximately 100 $cm^3$ of distilled water to the beaker and stir until all the solid has dissolved.
5. Pour the solution into a 250 $cm^3$ volumetric flask.
6. Add distilled water until the top of the meniscus is level with the graduation mark.

Suggest three improvements to this method.

**[3 marks]**

**07** An experiment is done to determine the relative molecular mass of a solid acid, $H_2A$, by titration with 0.380 mol $dm^{-3}$ sodium hydroxide solution.

Table 2 shows the results of the experiment.

**Table 2**

| | Rough | 1 | 2 | 3 |
|---|---|---|---|---|
| Final burette reading / $cm^3$ | 11.05 | 23.40 | 34.25 | 44.80 |
| Initial burette reading / $cm^3$ | 0.00 | 12.75 | 23.40 | 34.25 |
| Titre / $cm^3$ | 11.05 | 10.65 | 10.85 | 10.55 |

**07.1** The uncertainty in the use of the burette in each titration is ±0.15 $cm^3$

Calculate the percentage uncertainty in the use of the burette in titration 1.

**[1 mark]**

**07.2** Suggest one way to reduce the percentage uncertainty in each titration without changing the apparatus.

Justify your answer.

**[2 marks]**

**08** A student does a titration using a pipette to add 25.0 $cm^3$ of 0.112 mol $dm^{-3}$ aqueous sodium hydroxide to a conical flask. This aqueous sodium hydroxide is titrated with a solution of an unknown diprotic acid, $H_2A$, from a burette.

**08.1** The uncertainty in using the pipette in this experiment is ±0.06 $cm^3$

Calculate the percentage uncertainty in using the pipette.

**[1 mark]**

**08.2** Before adding the solution from the burette in the rough titration, there was an air bubble below the tap.

At the end of this titration the air bubble was not there.

Explain why this air bubble increases the final burette reading of the rough titration.

**[1 mark]**

**08.3** During the titration the student washed the inside of the conical flask with some distilled water.

Suggest why this washing does not give an incorrect result.

**[1 mark]**

**09** The student rinses the burette with deionised water before filling with sodium hydroxide solution.

State and explain the effect, if any, that this rinsing will have on the value of the titre.

**[2 marks]**

# Mark Scheme

*M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.*

| Question | Marking guidance | Comments | Mark |
|---|---|---|---|
| 01 | **C** | Source: 7404/1 Jun 2023 Q10 | 1 |
| 02 | **A** | Source: 7404/1 Jun 2023 Q11 | 1 |
| 03 | 0.20 ÷ 250 × 100 = 0.080% | Source: 7404/1 Jun 2023 Q03.2 | 1 |
| 04 | Additional drops of solution could have entered the burette from the funnel (making the value on the burette lower) | Source: 7404/1 Nov 2020 Q02.2<br>Must imply that solution from funnel drips into burette | 1 |
| 05 | Less chance of splashing / losing any solution using a conical flask (when swirling) | Source: 7404/1 Nov 2020 Q02.3<br>Allow easier to swirl | 1 |
| 06 | ANY THREE:<br>Record all masses (accurately to 2 decimal places)<br>Weigh by difference / wash the solid from weighing container into the beaker / add solid directly to volumetric flask (via a funnel) and dissolve in approximately 100 $cm^3$ of distilled water<br>Wash the beaker into the flask after the solution is transferred / wash the stirring rod into the flask / wash beaker and transfer washings to the volumetric flask<br>(Use a dropper when adding close to the graduation mark to) ensure the bottom of the meniscus is on the graduation mark<br>Mix thoroughly the final solution in the volumetric flask / invert the flask several times | Source: 7404/1 Jun 2023 Q03.1<br>Ignore apparatus changes | 3 |
| 07.1 | 0.15 ÷ 10.65 × 100 = 1.4(1)% | Source: 7404/1 Jun 2025 Q05.2–05.3 | 1 |
| 07.2 | M1 Use more dilute NaOH / use a larger mass of solid $H_2A$ / use a larger concentration of $H_2A$<br>M2 Would lead to a larger titre value / larger volume of NaOH required | | 2 |
| 08.1 | 0.06 ÷ 25.0 × 100 = 0.24% | Source: 7404/1 Jun 2019 Q06.2–06.4 | 1 |
| 08.2 | Some solution / acid replaces air bubble / solution / acid fills below the tap / air bubble takes up volume that would be filled by solution / acid | Allow acid / solution fills the bubble / gap<br>'The final reading is higher than the volume added' is not enough | 1 |
| 08.3 | Does not react (with the alkali) / does not change the number of moles (of alkali) | Allow water is a product / water is not a reagent | 1 |
| 09 | M1 Titre value would increase / larger value<br>M2 Because the sodium hydroxide solution would be more dilute | Source: 7404/1 Nov 2021 Q08.2 | 2 |